

PEIHUAN NI

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Institute of Automation, Chinese Academy of Sciences

95 Zhongguancun East Road, Haidian District, Beijing, China

EDUCATION

University of Chinese Academy of Sciences

Sep. 2024 – Jun. 2027 (expected)

Master's student in Artificial Intelligence, School of Artificial Intelligence

Advisors: Jian Cheng and Gang Li

GPA: 3.81

Jilin University

Sep. 2020 – Jun. 2024

Bachelor's degree in Microelectronics Science and Engineering

College of Electronic Science and Engineering

Advisor: Kaige Yan

College English Test: CET-4: 603; CET-6: 553

RESEARCH EXPERIENCE

First-Author Publications

Peihuan Ni, Zitao Mo, Tielong Liu, Hongli Wen, Zeyu Zhu, Minnan Pei, Junwen Si, Weifan Guan, Peisong Wang, Qinghao Hu, Gang Li and Jian Cheng. **APEX: Integer-only Non-linear Function Approximation for Efficient Cross-Modal Inference**. In **DATE 2026**. (CCF-B)

- Identified substantial differences in quantized data distributions across model modalities, leading to varying sensitivity to nonlinear approximation errors; retaining high precision uniformly incurs excessive computational bit widths and hardware costs.
- Proposed a unified integer-only architecture that merges the computation graphs of common nonlinear operators and uses bit-level pruning to reduce bit widths while preserving model accuracy.
- Improved task accuracy by up to approximately 0.7% for language models and 1.3% for vision models, while enabling near-lossless inference for multimodal models.
- Achieved area reduction factors of $1.73\text{--}8.71\times$ and power reduction factors of $1.21\text{--}10.83\times$ relative to existing methods.

Peihuan Ni, Zeyu Zhu, Weifan Guan, Gang Li, Zitao Mo, Tielong Liu, Junwen Si, Qinghao Hu, Peisong Wang and Jian Cheng. **ACE-VLA: Action-Centric Evaluation for Token Pruning in Vision-Language-Action Models**. In **CoRL 2026**. (BAAI Class A)

- Identified that existing VLA pruning methods largely adopt text-vision attention metrics from VLMs, underusing task-phase and motion semantics encoded in action outputs.
- Proposed an action-centric gradient-based importance metric and a motion-trajectory-based dynamic token pool to reduce importance evaluation overhead.
- Modeled robot motion trends from a kinematic perspective to dynamically adjust pruning ratios beyond instantaneous motion states.
- Improved task success rates by approximately 1.4% on $\pi_{0.5}$ and 2.2% on π_0 , with $1.33\text{--}1.57\times$ inference speedups; outperformed existing baselines in overall performance on OpenVLA-OFT.
- Validated transferability through real-robot experiments and improved the interpretability of VLA token pruning.

Co-Authored Publications

Zeyu Zhu, Gang Li, Minnan Pei, Zitao Mo, **Peihuan Ni**, Peisong Wang, Tielong Liu and Jian Cheng. **KL-MoE: A Hierarchical MoE Pruning Framework Exploiting KL Divergence**. In **DAC 2026**. (CCF-A)

Tielong Liu, Zeyu Zhu, Gang Li, Zitao Mo, Chen Bai, Junwen Si, **Peihuan Ni**, Minnan Pei, Peisong Wang, Zhuoran Song, Xiaoyao Liang and Jian Cheng. **FLUTE: A LUT-Based Floating-Point Engine for Efficient LLM Acceleration Exploiting Core-Tail Hierarchical Alignment**. In **MICRO 2026**. (CCF-A)

Junwen Si, Gang Li, Tielong Liu, **Peihuan Ni**, Minnan Pei, Zeyu Zhu, Zitao Mo, Peisong Wang, Zhongfeng Wang and Jian Cheng. **DOLA: A Dataflow Optimization Framework for Efficient Inference of Linear Attention Models on FPGAs**. Submitted to **TCAD**. (CCF-A)

Zeyu Zhu, Gang Li, Peisong Wang, **Peihuan Ni**, Tielong Liu, Minnan Pei, Junwen Si, Zhuoran Song, Xiaoyao Liang and Jian Cheng. **DALI: A Workload-Aware Offloading Framework for Efficient MoE Inference on Local PCs**. Submitted to **ATC 2026**. (CCF-A)

INTERNSHIP EXPERIENCE

Huawei Technologies Co., Ltd., Computing Product Line

Jul. 2026 – Jul. 2027

LLM Inference System Acceleration on the Ascend 950 NPU

- Analyze and optimize LLM inference on the Ascend 950 NPU, focusing on operator scheduling, on-chip dataflow, and compute resource utilization.
- Investigate bottlenecks across Cube and Vector units and the memory hierarchy, and explore system-level optimizations for MoE and long-context inference.

PROJECTS

VLA Model Quantization and FPGA Edge Deployment

Project Lead

- Designed a W4A8 quantization scheme with 4-bit weights and 8-bit activations and KV cache, combined with integer-only nonlinear approximations for near-lossless edge inference.
- Implemented fused quantization and dequantization operators using FP16-INT8 multiply-accumulate (MAC) units, reducing intermediate writebacks and off-chip memory accesses while keeping more activations on chip.
- Reused FP16-INT8 MAC units for operators such as RoPE and reordered selected computations to improve compute-unit and on-chip data reuse.
- Designed bit-serial compute units for W4A8 mixed-precision matrix multiplication, a KV reordering module supporting different attention head dimensions, and crossbar-based data mapping from BRAM to the compute array.
- Designed a unified integer-only nonlinear unit supporting multiple operators with reduced hardware area and power consumption; results published at DATE 2026.
- Contributed to VLIW compiler design, with responsibility for on-chip dataflow, the instruction processor, and operator control modules; reduced the instruction count per model inference by over $32\times$ using address offsets and multiple offset modes.
- Built a GPU-based bit-accurate simulator and verified consistency with Vivado module-level simulations and full-model execution on FPGA.
- Completed end-to-end FPGA deployment, including DMA memory-access debugging, HBM bandwidth optimization, VH158 board data-path debugging, timing closure, and PetaLinux system builds.

LLM Analysis and Inference Optimization on the Ascend A5 NPU

Project Lead

- Analyzed bottlenecks caused by asymmetric throughput between matrix and vector compute units in the Ascend architecture, identifying nonlinear operator overhead, data movement, and idle compute resources in MoE inference.
- Explored fine-grained pipelining and overlapping of matrix multiplication and nonlinear computation to balance Cube and Vector workloads and improve overall hardware utilization.

HONORS AND AWARDS

Outstanding Student, University of Chinese Academy of Sciences

Third Prize, Intel Cup National Undergraduate Electronic Design Contest – Invitational

Second Prize, China Undergraduate Mathematical Contest in Modeling

Third Prize, Chinese Mathematics Competitions for College Students

ACADEMIC SERVICE

Reviewer for WACV, EMNLP, and CoRL